

Mors certa, hora incerta.



Tag 14

09.07.2026

Donnerstag

Von KI erstellt, von Menschen kontrolliert.

Guten Morgen!

QED

Quantum Electro Dynamics
1963

Richard Feynman

— ~~~~~

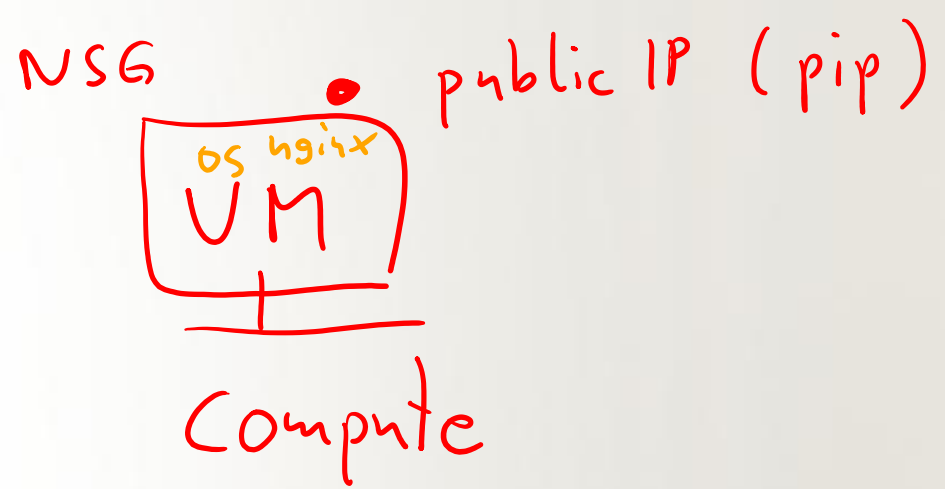
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GitHub Repo <https://github.com/www42/a5>

# AZ-104

## Learning Paths:

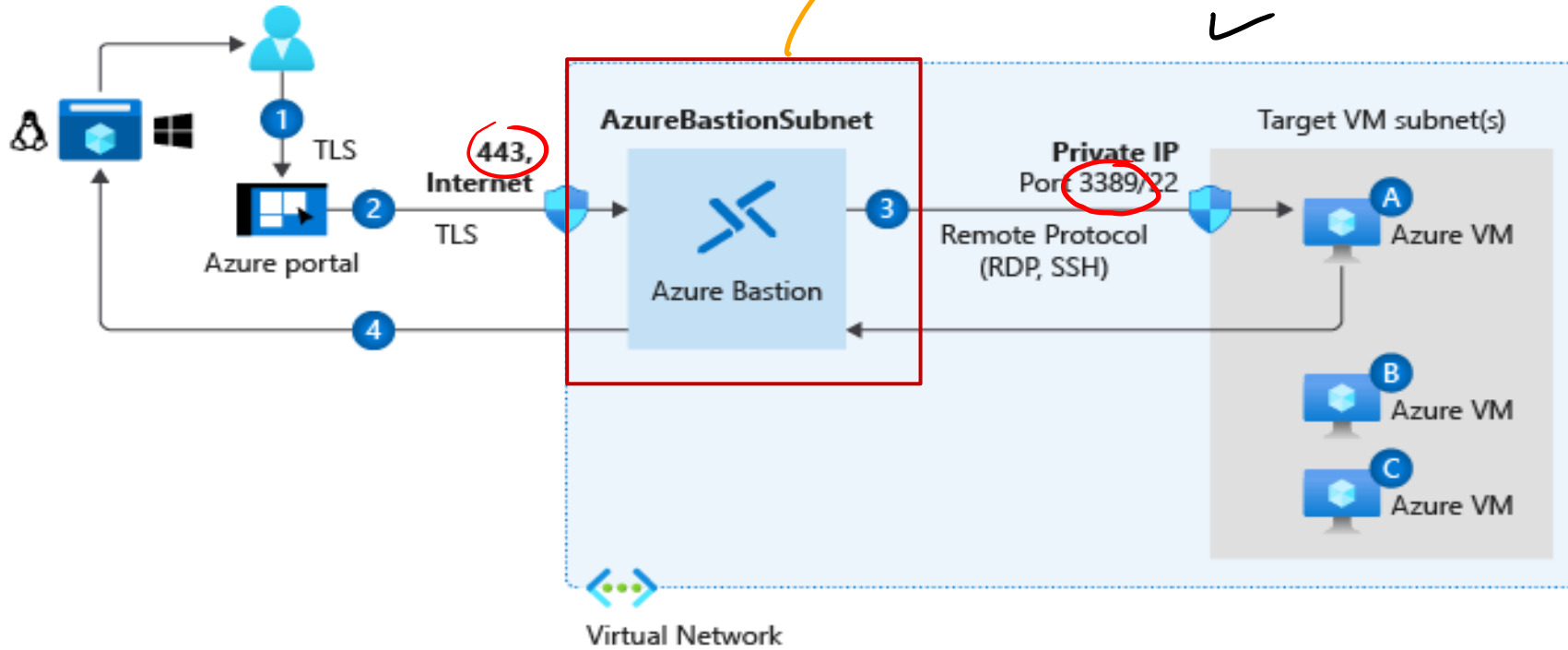
- Manage identities and governance in Azure
- Prerequisites for Azure administrators – CloudShell and templates
- Configure and manage virtual networks for Azure administrators
- Implement and manage storage in Azure
- Deploy and manage Azure compute resources
- Monitor and back up Azure resources



# What happened yesterday

# Connect to Virtual Machines

Student SKU (Free)  
(Developer!) Preview



Bastion Subnet for RDP/SSH through the Portal over SSL

Remote Desktop Protocol for Windows-based Virtual Machines

Secure Shell Protocol for Linux based Virtual Machines

# Connect to Windows Virtual Machines

NSG

Remote Desktop Protocol (RDP)  
creates a GUI session and accepts  
inbound traffic on TCP port 3389

Windows Remote Management (WS-Man)

WinRM creates a command-line  
session so you can run scripts

5985

dir wsman:



Tolle Filme: "23"  
"Der Dritte Mann"

# Connect to Linux Virtual Machines

Administrator account

Authentication type

Username \* ⓘ

SSH public key \* ⓘ

Provide an RSA public key in the single-line format (starting with "ssh-rsa") or the multi-line PEM format. You can generate SSH keys using ssh-keygen on Linux and OS X, or PuTTYGen on Windows.



NSG

22

[Learn more about creating and using SSH keys in Azure](#)

Authenticate with a SSH public key or password

SSH is an encrypted connection protocol that allows secure logins over unsecured connections

There are public and private keys

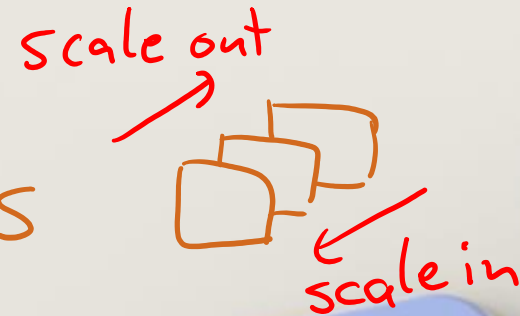
# Heute

- Docker

(flexible)  
→ Uniform

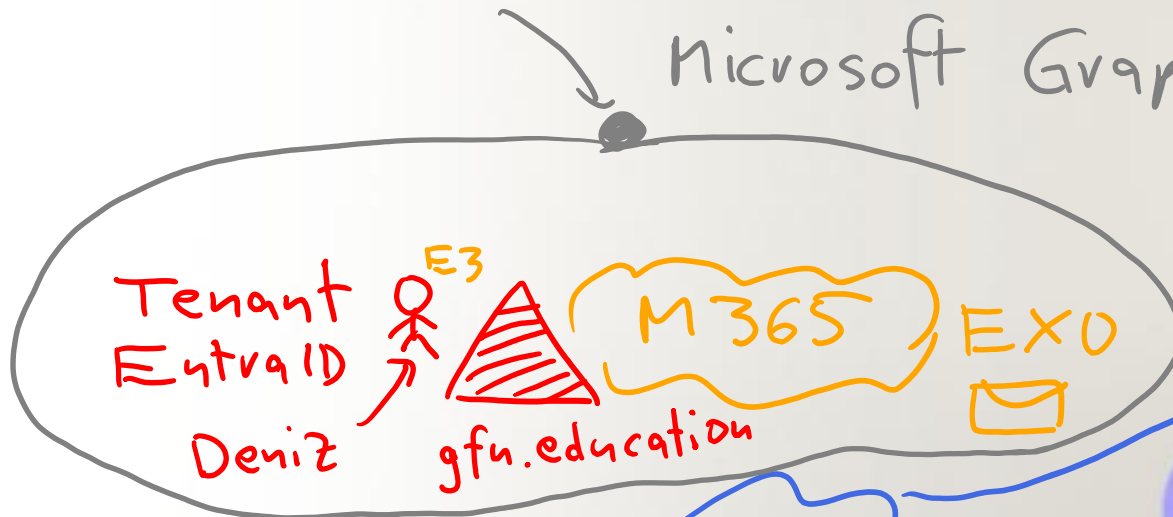
- Avail. Set VMs
- Scale Set VMs

VMSS



- Skillable Lab 08

Microsoft Graph API



[atropos.onmicrosoft.com](https://atropos.onmicrosoft.com)

Azure

ARM API

Microsoft

Powerpoint  
Excel  
Word

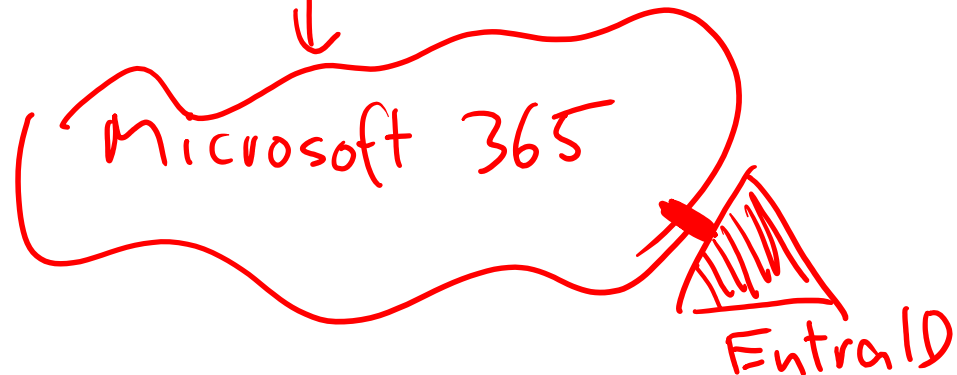
windows 95

NT

windows Server 2000



Office 365

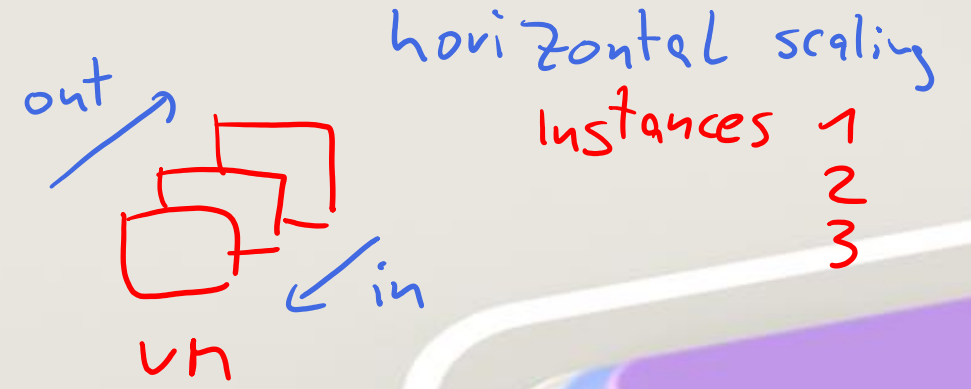
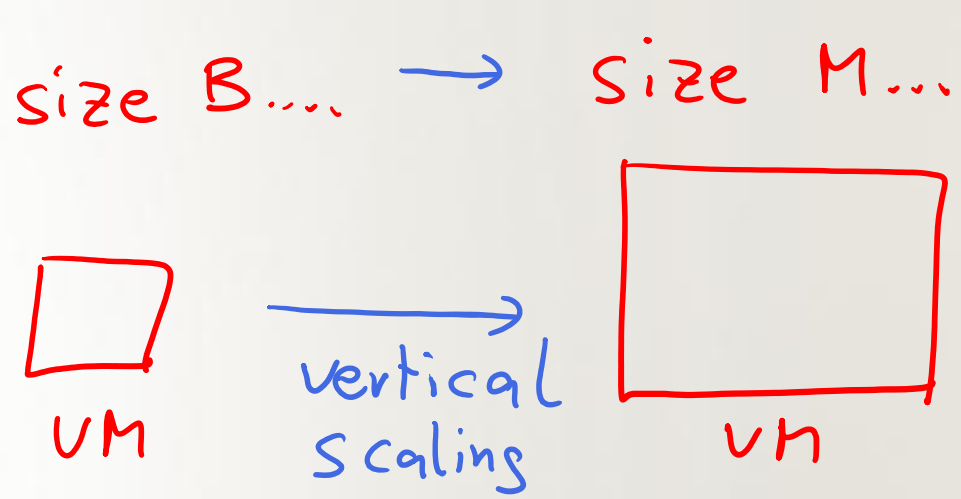






# AZ-104

 Deploy and manage  
Azure compute resources



# Configure Virtual Machine Availability

# Setup Availability Sets

## Instance details

Name \* ⓘ

avset01



Region \* ⓘ

(US) East US



Fault domains ⓘ



2

Update domains ⓘ



5

Use managed disks ⓘ

No (Classic)

Yes (Aligned)

Two or more instances in  
Availability Sets = 99.95% SLA

Configure multiple  
Virtual Machines in  
an Availability Set

Configure each  
application tier  
into separate  
Availability Sets

Combine a Load  
Balancer with  
Availability Sets

Use managed disks  
with the Virtual  
Machines

# VM Avail. Set

|                    | Fault Domain<br>0 | Fault Domain<br>1 | 2   |
|--------------------|-------------------|-------------------|-----|
| Update Domain<br>0 | VM1               |                   |     |
| 1                  | VM4               | VM2               |     |
| 2                  |                   | VM5               | VM3 |
|                    |                   |                   |     |

# Review Availability Zones

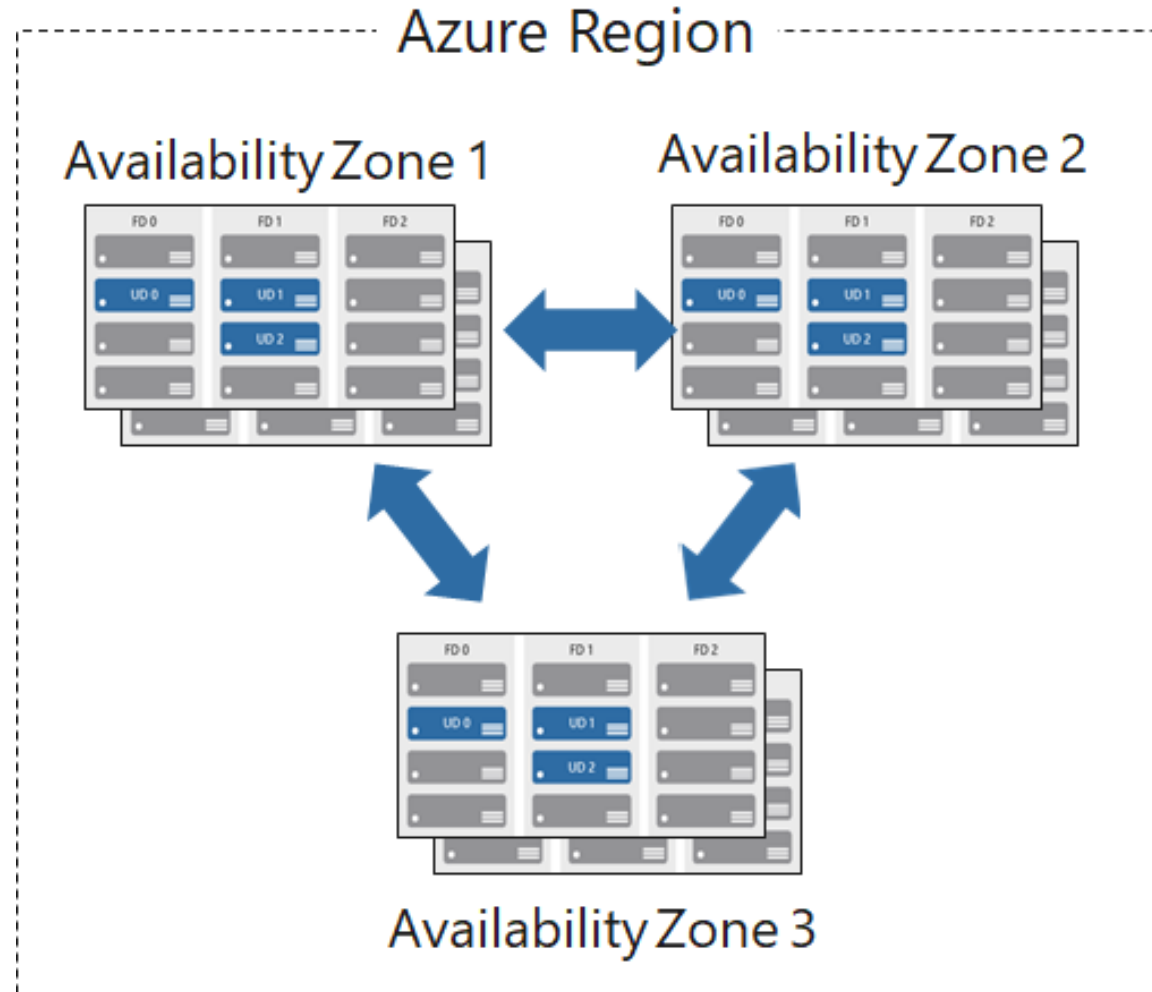
Unique physical locations in a region

Includes datacenters with independent power, cooling, and networking

Protects from datacenter failures

Combines update and fault domains

Provides 99.99% SLA



# Compare Vertical to Horizontal Scaling

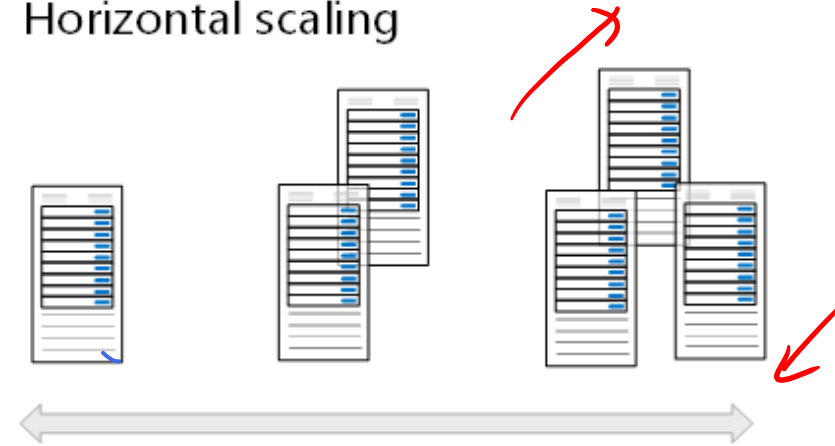
VMSS (Virtual Scale Set)

Vertical scaling



Vertical scaling (scale up and scale down) is the process of increasing or decreasing power to a single instance of a workload; usually manual

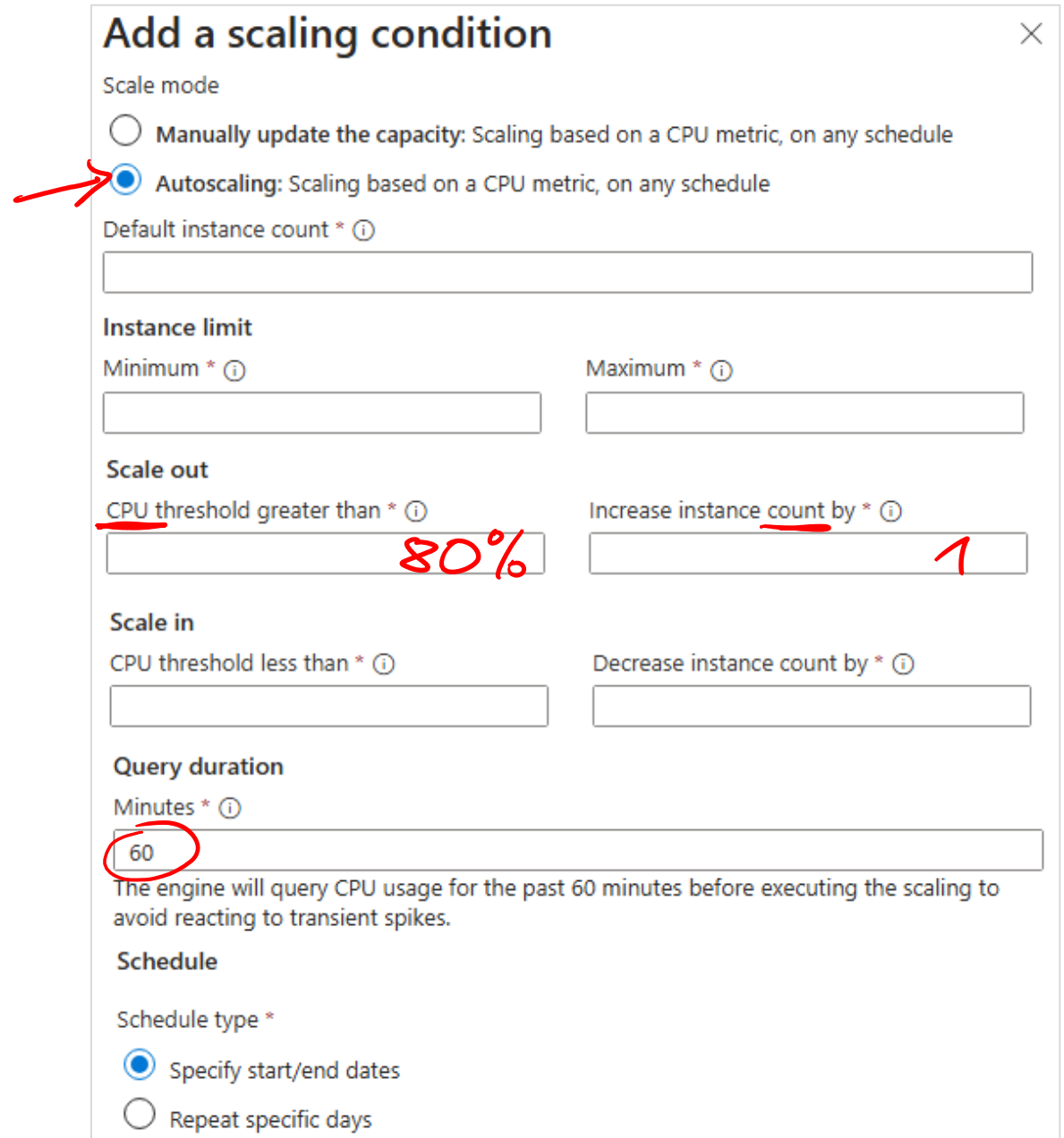
Horizontal scaling



Horizontal scaling (scale out and scale in) is the process of increasing or decreasing the number of instances of a workload; frequently automated

# Create and configure scaling

- Manually update or autoscale
- Define a minimum, maximum, and default number of VM instances
- Create more advanced scale sets with scale out and scale in parameters



**Add a scaling condition** [X]

Scale mode

☐ Manually update the capacity: Scaling based on a CPU metric, on any schedule

☒ Autoscaling: Scaling based on a CPU metric, on any schedule

Default instance count \* ⓘ

**Instance limit**

Minimum \* ⓘ  Maximum \* ⓘ

**Scale out**

CPU threshold greater than \* ⓘ  Increase instance count by \* ⓘ

**Scale in**

CPU threshold less than \* ⓘ  Decrease instance count by \* ⓘ

**Query duration**

Minutes \* ⓘ

The engine will query CPU usage for the past 60 minutes before executing the scaling to avoid reacting to transient spikes.

**Schedule**

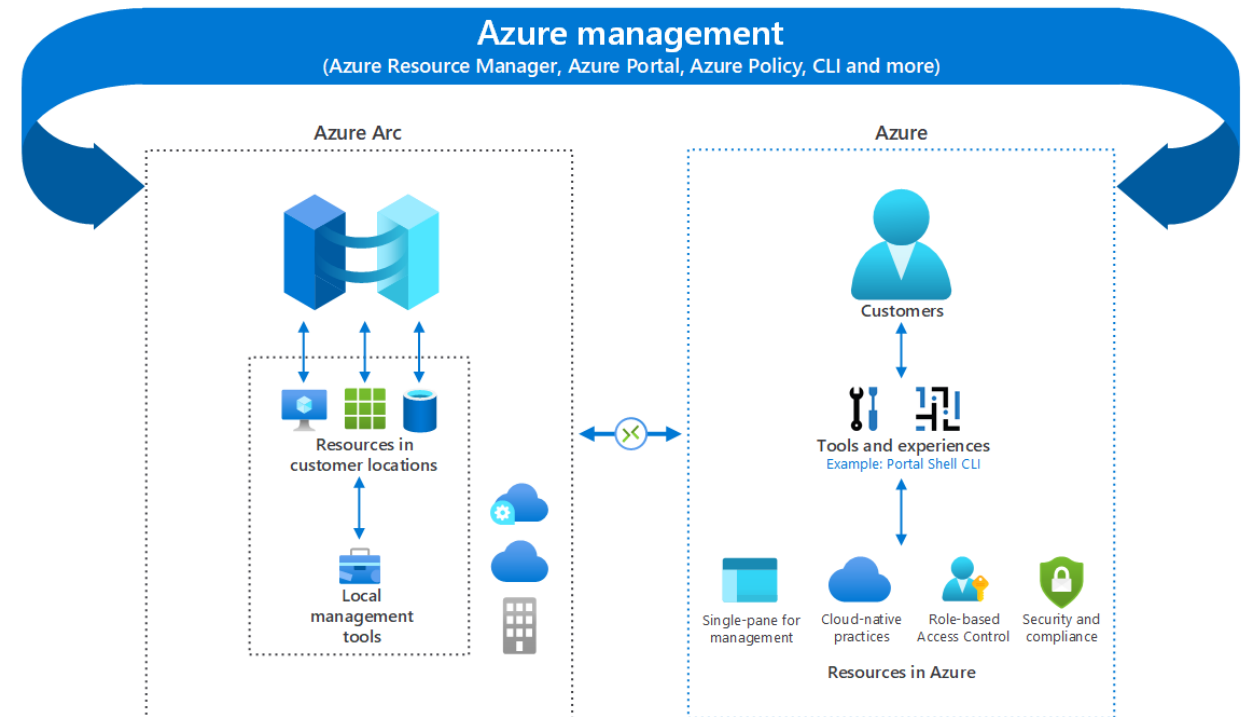
Schedule type \*

☒ Specify start/end dates

☐ Repeat specific days

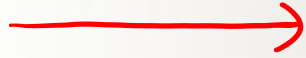
# Bring Azure innovation to your hybrid environments with Azure Arc

- Manage your entire environment of existing non-Azure and/or on-premises resources
- Manage virtual machines, Kubernetes clusters, and databases as if they are running in Azure.
- Use familiar Azure services and management capabilities, regardless of where your resources live
- Use traditional IT Ops while introducing DevOps practices to support new cloud native patterns in your environment





VM



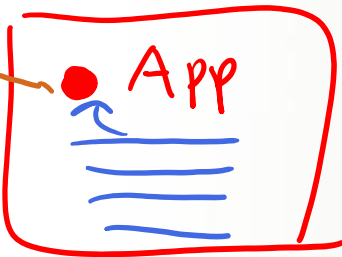
Docker  
Container

Cloud Native Apps

And now for something  
completely different...

Container

process



Image

App

.zip .rar  
||| |||

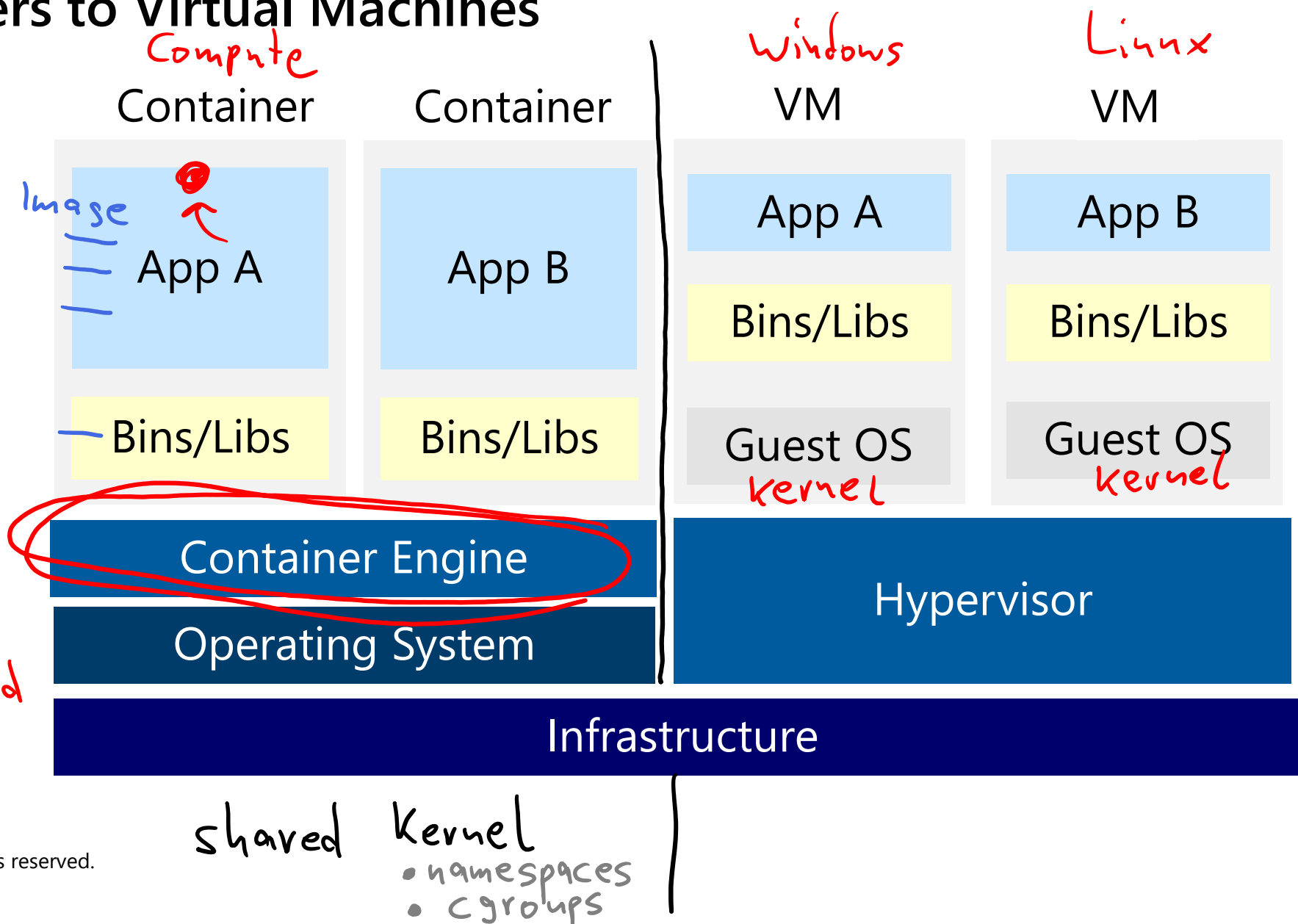
Container = Image, bei dem  
die App gestartet wurde

File System  
(unionfs)  
Standard Format

# Compare Containers to Virtual Machines

- Isolation
- Operating System
- Deployment
- Persistent storage
- Fault tolerance

*dockerd  
containerd*



Docker Registry

Docker Image

Overhead  
Projektor  
Folien



Layer

z.B.  
Docker Hub



App



Lib



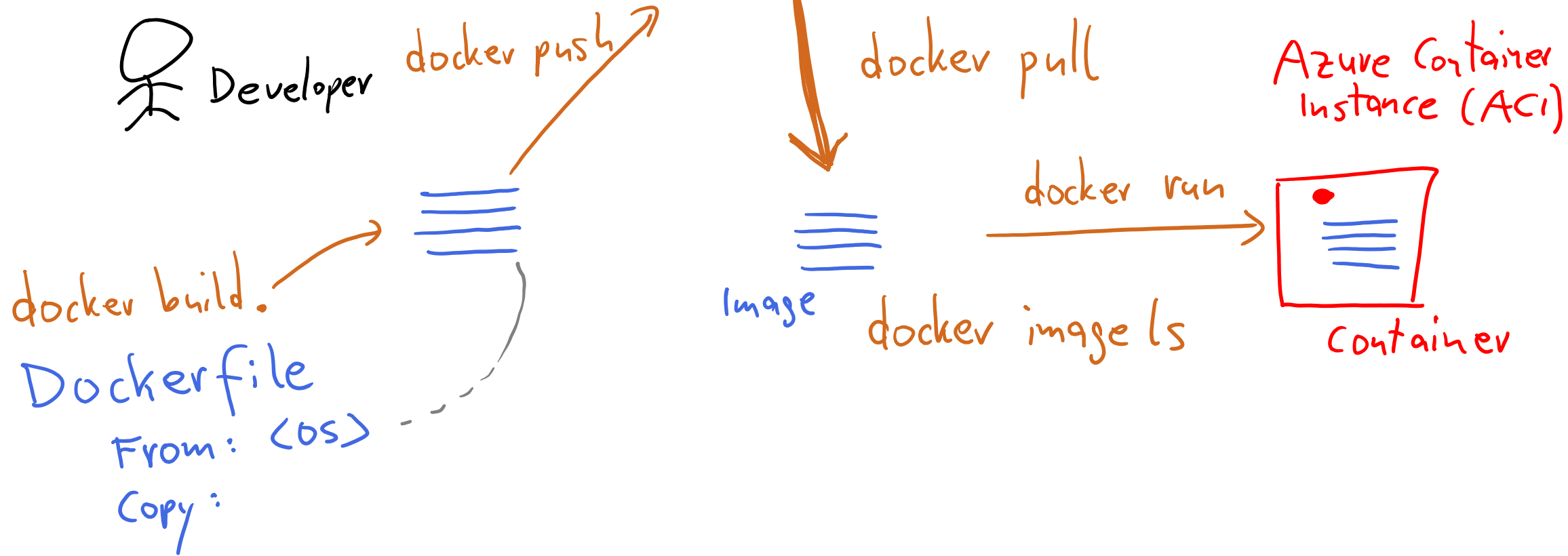
OS (Linux abgespeckt)

Docker Images sind  
Read Only!

# Quickstart

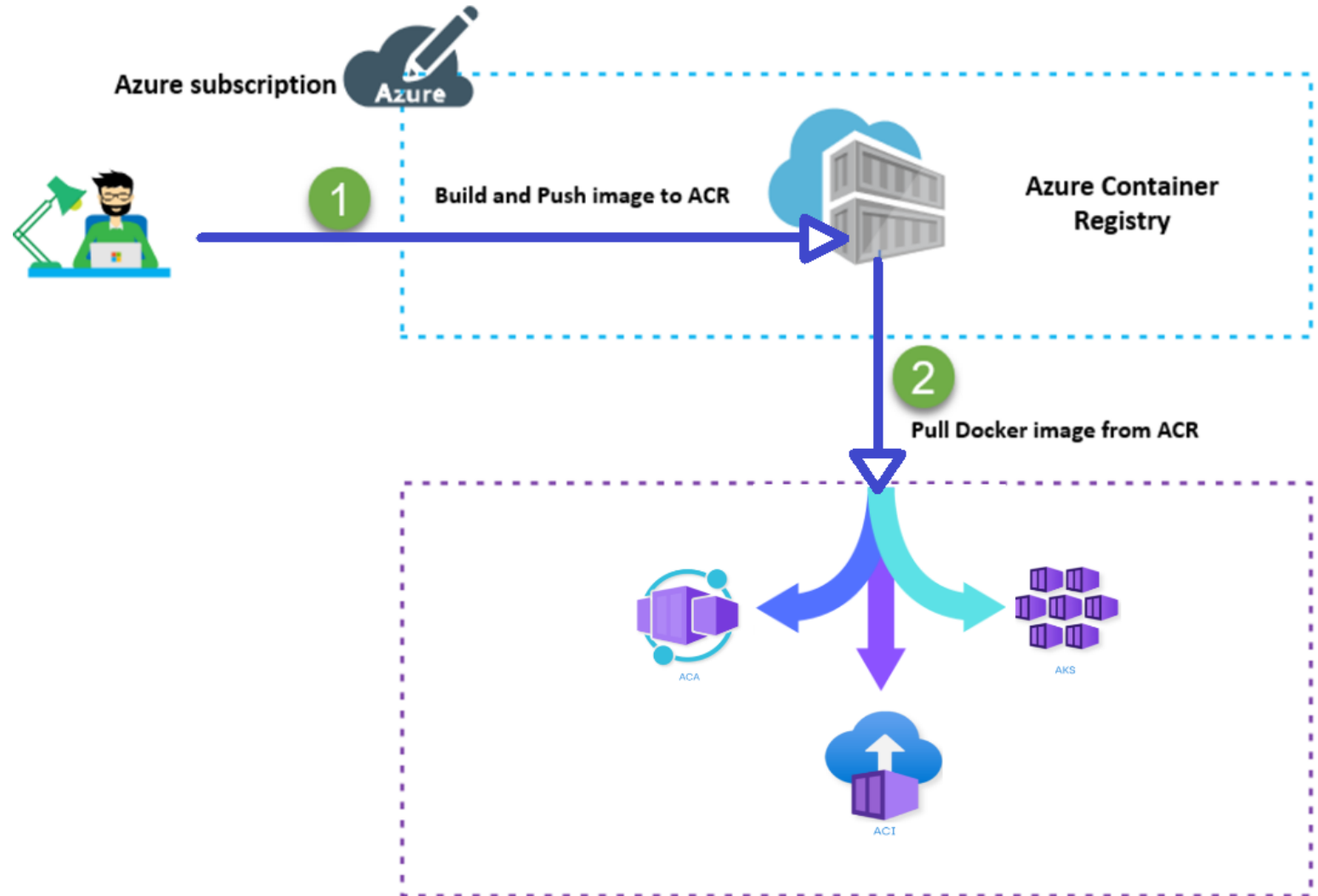
Registry

Docker Hub  
Azure Container Registry (ACR)



# Understand Container Images

A container image is a lightweight, standalone, executable package of software that encapsulates everything needed to run an application.



# Labs





TO THE LAB ZONE →

NO THROUGH ROUTE

Berlin  
Europa  
Dublin

SPEEDY ROUTE

4<sup>2</sup>

lounge

Microsoft  
tech-ed  
2010

lounge



software diagnostics

E94

SD Studio

for Team Foundation Server  
Visualize your code in  
a software map.

SD Developer Edit

for C and C++ for Visual  
Trace dynamically your code  
Visualize the control flow.

software  
diagnostics

DataCore  
SOFTWARE

Skillable AZ-104

## Lab 08 – Manage Virtual Machines

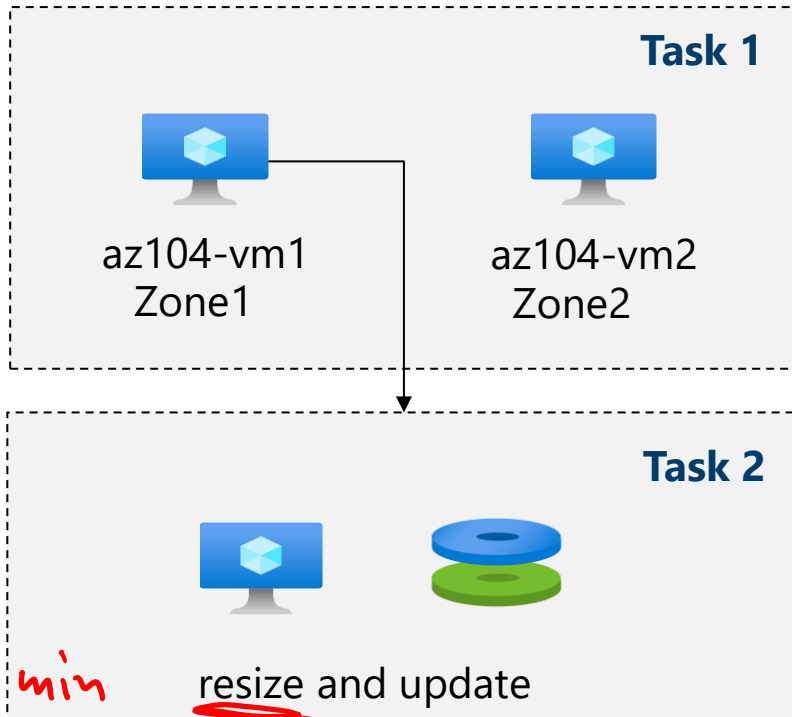




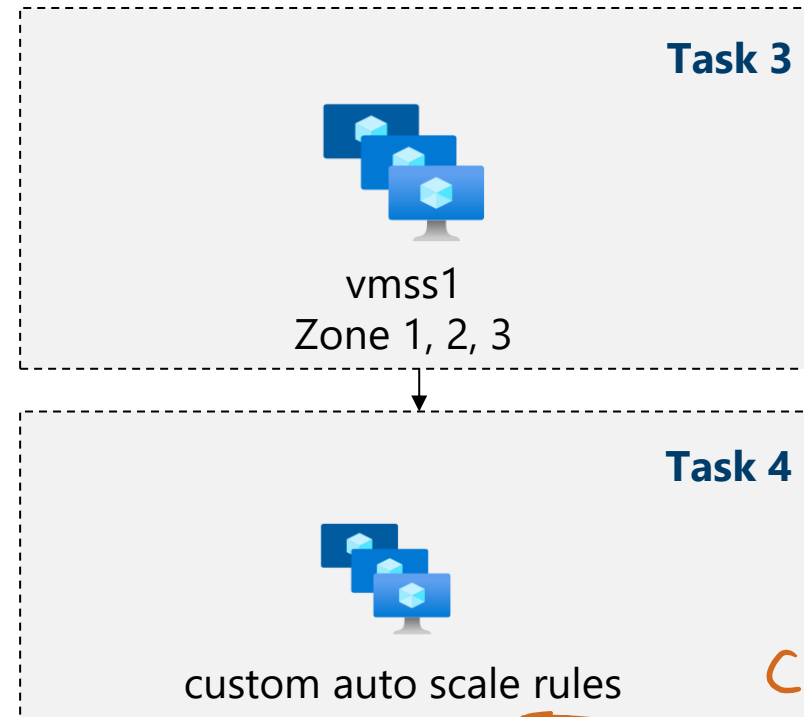
# Lab 08 – Architecture diagram

1.  
vertical

## Azure Virtual Machines



## Virtual Machine Scale Sets



2.  
horizontal

**Task 5:** Create a virtual machine using Azure PowerShell (option 1)

**Task 6:** Create a virtual machine using the CLI (option 2)

`az vm create ...`

`New-AzVm ...`

# End of presentation



On Prem  
GFW Remote Lab

Hyper-V

Virtual  
Appliance

VM 1

virtual switch

Azure

Recovery Service Vault

Backup  
ASR  
Migration

Register  
PSK

Migrate

:443



vNet

